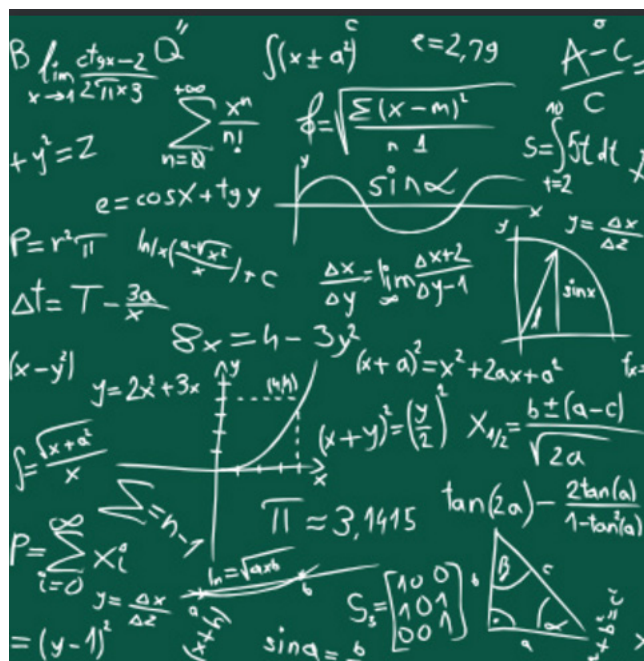


Corner points, cusps, points with vertical/horizontal tangent



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani e Sandro Salsa, Zanichelli.

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1 Right derivative and left derivative

Definition 1: of right (left) derivative

Let $f : (a, b) \rightarrow \mathbb{R}$; the function f is said to be differentiable at $x_0 \in (a, b)$ from the right (from the left) if the following limit exists and is finite

$$\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} \quad \left(\lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h} \right)$$

then f is differentiable from the right (from the left) and the limit is called the **right derivative (left derivative)**.

- The right derivative is denoted by the symbol $f'_+(x_0)$, while the left derivative is denoted by the symbol $f'_-(x_0)$

2 Corner points

Definition 2: of corner point

If f is continuous and differentiable from the right and from the left (but not differentiable) at x_0 , we say that f has a **corner point** at $x = x_0$.

Example 1: Corner points

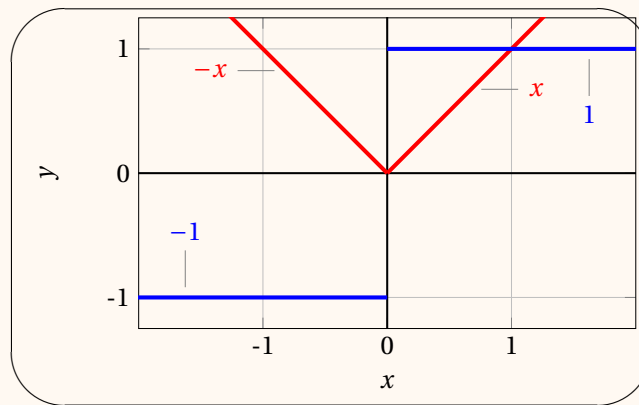
Consider the function $f(x) = |x|$ and the point $x_0 = 0$. We have:

$$f'_+(0) = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1 \quad \text{and} \quad f'_-(0) = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

Since the limit of the difference quotient does not exist, f is not differentiable at $x = 0$. The function is continuous at $x = 0$ (the origin) since

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -x = 0 \quad \text{hence} \quad \lim_{x \rightarrow 0} f(x) = f(0) = 0$$

Since the right and left limits of the difference quotient at $x = 0$ exist and are finite, the graph therefore has a corner point at $x = 0$.



The formula that concisely expresses the derivative of the absolute value function (away from the origin) is the following:

$$f(x) = |x|, \quad f'(x) = \operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

3 Points with vertical/horizontal tangent

- If f is continuous at a point x_0 and

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \pm\infty$$

then f is not differentiable at x_0 but, geometrically, the graph of f has a well-defined tangent line parallel to the y -axis.

In this case we will allow the notation

$$f'(x_0) = +\infty, \quad f'(x_0) = -\infty$$

and we will speak of a **point with vertical tangent**.

Definition 3: of point with vertical tangent

If

$$f'(x_0) = +\infty \quad \text{or} \quad f'(x_0) = -\infty$$

*we say that f has a **point with vertical tangent** at $x = x_0$.*

- Similarly, if f is continuous at a point x_0 and

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = 0$$

the graph of f has a well-defined tangent line parallel to the x -axis. In this case we will speak of a **point with horizontal tangent**.

Definition 4: of point with horizontal tangent

If

$$f'(x_0) = 0$$

*we say that f has a **point with horizontal tangent** at $x = x_0$.*

Example 2: Point with vertical tangent

Let

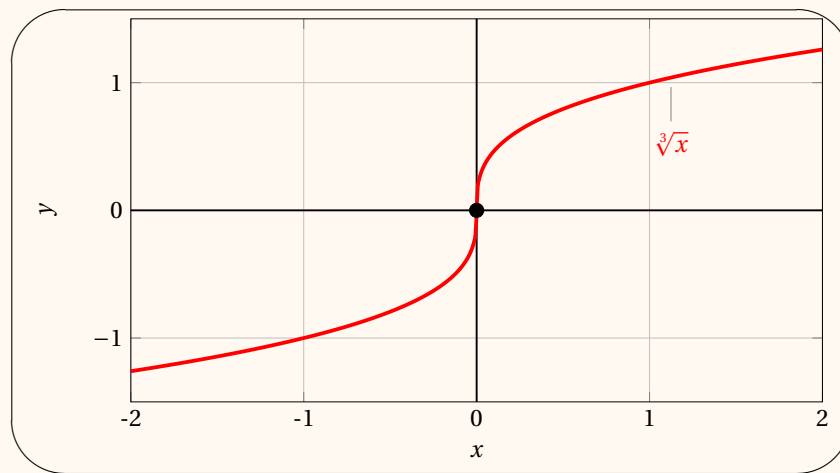
$$f(x) = \sqrt[3]{x}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{\sqrt[3]{h}}{h} = \frac{1}{h^{2/3}}$$

and the limits are:

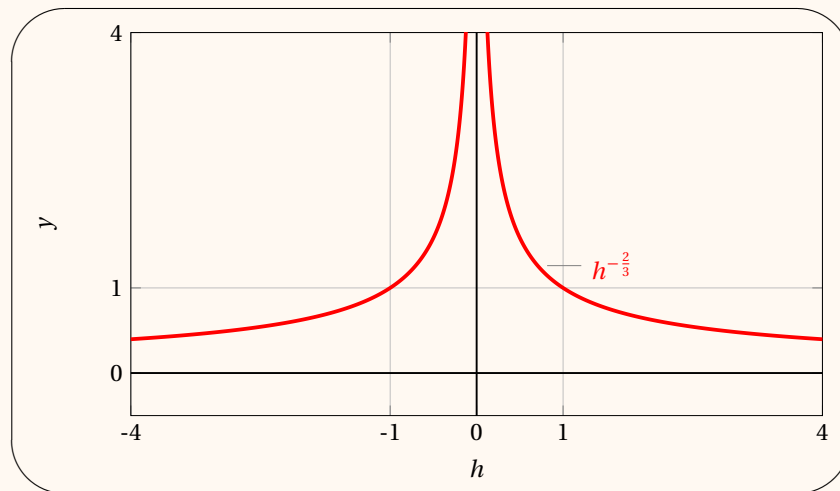
$$\lim_{h \rightarrow 0^-} \frac{1}{h^{2/3}} = +\infty \text{ and } \lim_{h \rightarrow 0^+} \frac{1}{h^{2/3}} = +\infty \text{ hence } \lim_{h \rightarrow 0} \frac{1}{h^{2/3}} = +\infty \text{ and } f'(0) = +\infty.$$



The function has a point with vertical tangent at $x_0 = 0$. The function:

$$f(h) = \frac{1}{h^{2/3}} = h^{-\frac{2}{3}}$$

is a power with negative rational exponent $\frac{m}{n}$ with n odd (hence defined on $\mathbb{R} \setminus \{0\}$) and m even (hence an even function). Its graph is:



Example 3: Point with vertical tangent

Let

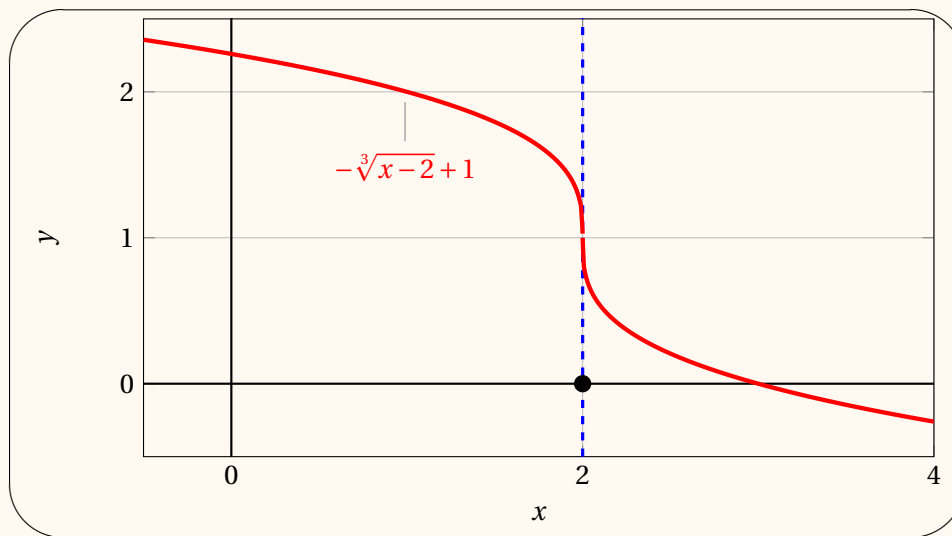
$$f(x) = -\sqrt[3]{x-2} + 1$$

then for $x_0 = 2$

$$\frac{f(2+h) - f(2)}{h} = \frac{-\sqrt[3]{(2+h)-2} + 1 - 1}{h} = -\frac{\sqrt[3]{h}}{h} = -\frac{1}{h^{2/3}}$$

and the limits are

$$\lim_{h \rightarrow 0^-} -\frac{1}{h^{2/3}} = -\infty \quad \text{and} \quad \lim_{h \rightarrow 0^+} -\frac{1}{h^{2/3}} = -\infty \quad \text{hence} \quad \lim_{h \rightarrow 0} -\frac{1}{h^{2/3}} = -\infty \quad \text{and} \quad f'(2) = -\infty.$$



The function has a point with vertical tangent at $x_0 = 2$.

4 Cusps

Definition 5: of cusp

Let f be a function continuous at x_0 ; if

$$f'_+(x_0) = \pm\infty \quad \text{and} \quad f'_-(x_0) = \mp\infty$$

we say that f has a cusp at x_0 .

Example 4: Cusp

Let

$$f(x) = \sqrt[3]{|x|}$$

then for $x_0 = 0$

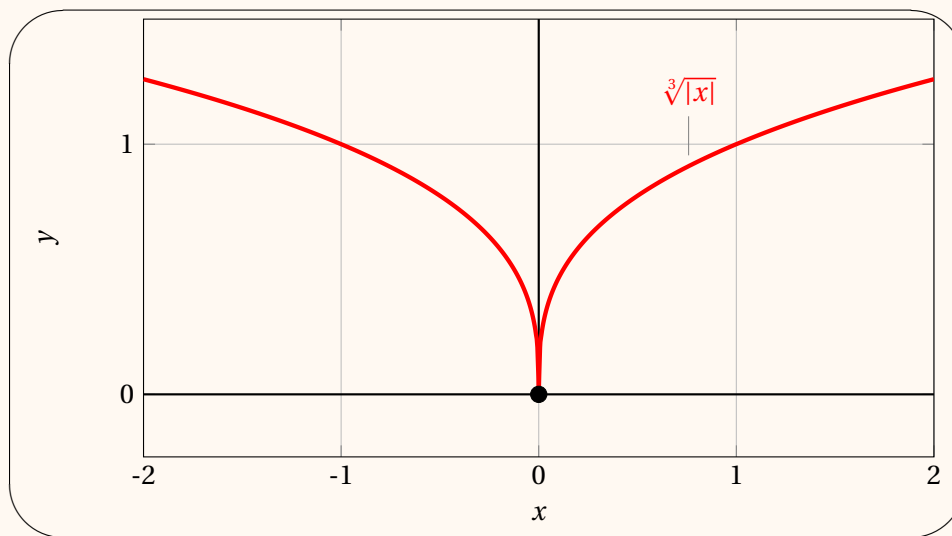
$$\frac{f(h) - f(0)}{h} = \frac{\sqrt[3]{|h|}}{h}$$

and the limits are:

$$\lim_{h \rightarrow 0^-} \frac{\sqrt[3]{-h}}{h} = \lim_{h \rightarrow 0^-} -\frac{1}{h^{2/3}} = -\infty \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{\sqrt[3]{h}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{h^{2/3}} = +\infty,$$

hence:

$$f'_-(0) = -\infty \quad \text{and} \quad f'_+(0) = +\infty.$$



The function has a cusp at $x_0 = 0$.

- In the mixed case where one of the two derivatives is finite and the other is infinite (with f continuous) we still speak of a corner point.
- Finally, if the function is defined only for $x \geq x_0$ and at that point has an infinite (right) derivative, we will simply say that at that point it has a vertical tangent, without speaking of either a cusp or a point with vertical tangent.

Example 5: Point with vertical tangent

Let

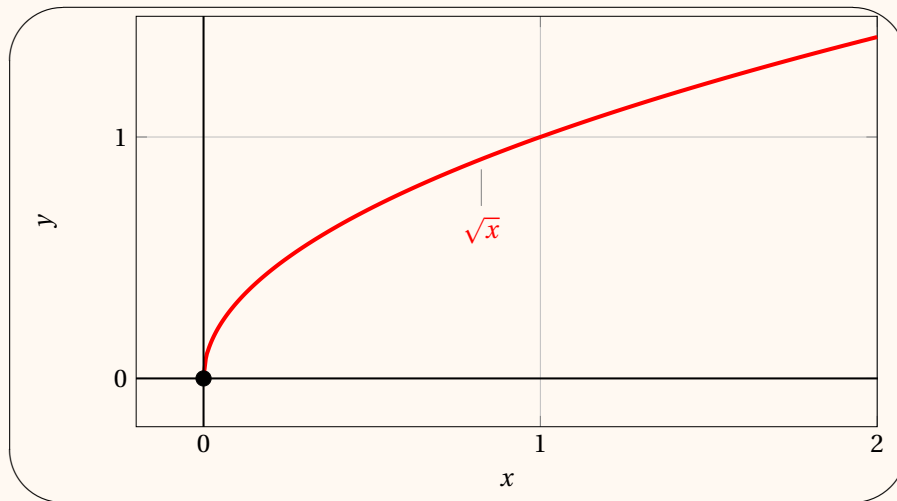
$$f(x) = \sqrt{x}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{\sqrt{h}}{h} = \frac{1}{h^{1/2}}$$

we only have the right limit

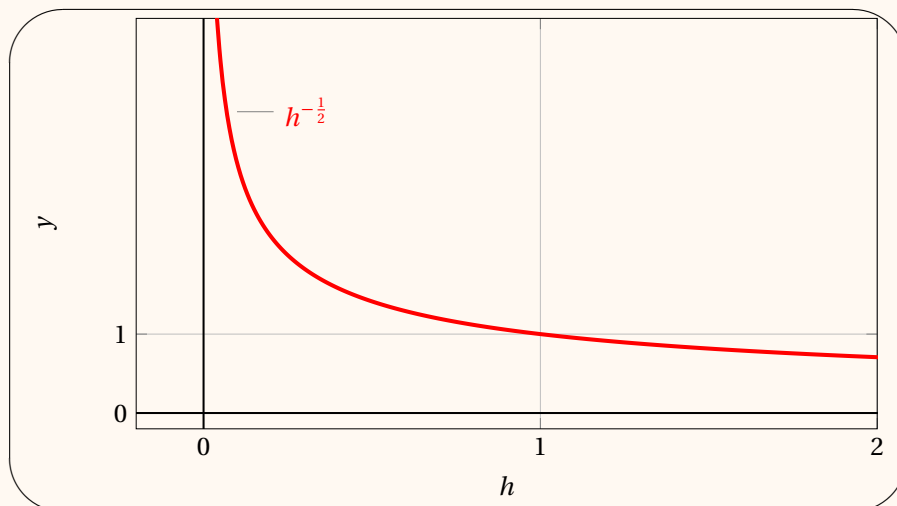
$$\lim_{h \rightarrow 0^+} \frac{1}{h^{1/2}} = +\infty \quad \text{and} \quad f'_+(0) = +\infty.$$



The function has a point with vertical tangent at $x_0 = 0$. The function:

$$f(h) = \frac{1}{h^{1/2}} = h^{-\frac{1}{2}}$$

is a power with negative rational exponent $\frac{m}{n}$ with n even (hence defined only on $\mathbb{R}_+$). Its graph is:



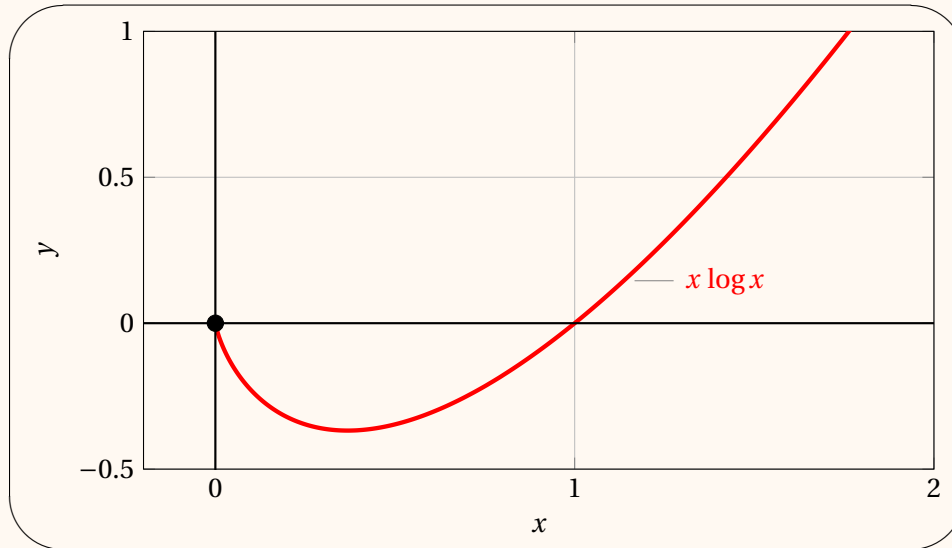
Example 6: Continuous extension from the right and behavior at the origin

Let

$$f(x) = x \log x, \text{ for } x > 0 \text{ we have } \lim_{x \rightarrow 0^+} x \log x = 0$$

hence the function can be extended by continuity from the right at $x = 0$, by setting:

$$f(x) = \begin{cases} x \log x & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$$



We compute the right derivative at $x_0 = 0$ of $f(x) = x \log x$ extended by continuity from the right at $x = 0$. We have, for $x_0 = 0$:

$$\frac{f(x_0 + h) - \overbrace{f(x_0)}^{=0}}{h} = \frac{f(x_0 + h)}{h} = \frac{f(h)}{h}$$

and consequently

$$\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{h \log h}{h} = \lim_{h \rightarrow 0^+} \log h = -\infty \quad \text{hence} \quad f'_+(x_0) = -\infty$$

The function therefore has a point with vertical tangent at $x_0 = 0$.

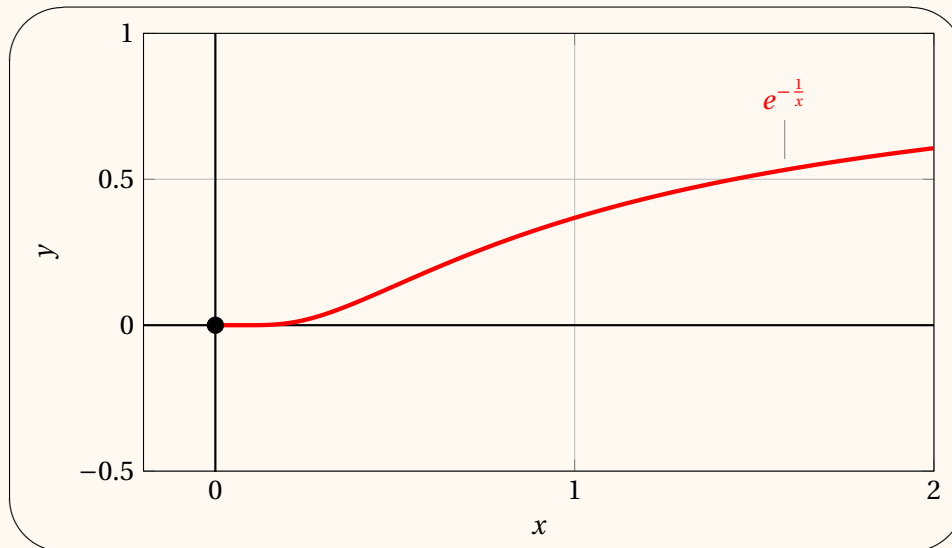
Example 7: Continuous extension from the right and behavior at the origin

Let

$$f(x) = e^{-\frac{1}{x}}, \text{ for } x > 0 \text{ we have } \lim_{x \rightarrow 0^+} e^{-\frac{1}{x}} = 0$$

hence the function can be extended by continuity from the right at $x = 0$, by setting:

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$$



We compute the right derivative at $x_0 = 0$ of $f(x) = e^{-\frac{1}{x}}$ extended by continuity from the right at $x = 0$.

We have:

$$\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{e^{-\frac{1}{h}}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{e^{\frac{1}{h}} h}$$

Changing the variable:

$$y = \frac{1}{h}, \text{ if } h \rightarrow 0^+ \text{ then } y \rightarrow +\infty$$

We obtain:

$$\lim_{h \rightarrow 0^+} \frac{1}{e^{\frac{1}{h}} h} = \lim_{y \rightarrow +\infty} \frac{y}{e^y} = 0 \text{ hence } f'_+(x_0) = 0$$

and the function has a point with horizontal tangent (from the right) at $x_0 = 0$.